

PRAJAS WADEKAR

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Education

International Institute of Information Technology (IIIT) | Hyderabad, India 2024 - 2028

B.Tech (Hons.) and M.Sc. in Computer Science (Dual Degree) CGPA: 7.95

- Coursework: Operating Systems and Networks, Design and Analysis of Software Systems, Statistical Methods in AI, Advanced NLP, Computer Vision, Information Extraction and Retrieval, Introduction to Information Security, Design and Analysis of Algorithms, Game Theory

Experience

Machine Learning Lab, IIIT Hyderabad May 2025 - Present

Undergraduate Researcher - Dr. Charu Sharma Hyderabad

- Spearheaded research on the adversarial robustness of 3D novel view synthesis methods, specifically targeting 3D and 4D Gaussian Splatting architectures.
- Formulated novel performance-degradation attacks against 3D reconstruction pipelines, establishing the first baseline vulnerability metrics for secure Gaussian Splatting evaluation.

Eklavya Foundation January 2025 - April 2025

Software Engineering Intern Hyderabad

- Lead a small development team to architect and deploy a custom Learning Management System (LMS) across the NGO's multiple institutions, supporting 2,000+ active students and faculty.
- Architected 15+ secure RESTful API endpoints and MongoDB schemas with Google OAuth and 3-tier RBAC, enabling concurrent access for 2,000+ users across role-specific dashboards.

Publications

PatchPoison: Poisoning Multi-View Datasets to Degrade 3D Reconstruction arXiv [📄](#) | [🔗](#)

P. Wadekar, V. S. P. Bachina*, K. Bhosikar*, A. Gangwal, C. Sharma | *CVPR Workshop on SPAR-3D, 2026*

- Proposed a lightweight adversarial patch framework targeting the feature-matching stage of Structure-from-Motion (SfM) pipelines (COLMAP) to prevent unauthorized 3D reconstruction.
- Demonstrated that injecting a minimal 12x12 pixel checkerboard patch increases reconstruction error (LPIPS) by 6.8x on the NeRF-Synthetic benchmark while remaining visually imperceptible.

Projects

FIRE: Fast Information Retrieval Engine | *Python, PostgreSQL, Redis, FastAPI, TF-IDF/BM25* [🔗](#)

- Engineered a full-text search engine over a 1M+ document corpus spanning Wikipedia and Webz.io data, building custom positional inverted indexes with TAAT/DAAT query execution for Boolean, phrase, and TF-IDF ranking.
- Accelerated query throughput by 99.1% (2,483 to 4,943 QPS) and cut p95 latency over 50% (to 0.31ms) by devising positional skip pointers and posting-list compression, benchmarked across Redis, PostgreSQL, and Elasticsearch.

ESPER : Insurance Claim Analysis System | *Python, Flask, YOLOv8, MySQL, CLIP* [🔗](#)

- Developed an automated AI claims pipeline, fine-tuning YOLOv8 across 17 vehicle-damage classes and fusing CLIP embeddings with Error Level Analysis (ELA) to flag fraudulent photos at 74% detection accuracy
- Deployed a scalable Flask and MySQL web application to process concurrent high-resolution image uploads, dynamically generating standardized PDF claim reports in under 3 seconds.

GeoVision: Geo-Localization from Campus Imagery | *PyTorch, ConvNeXt, ResNet, EfficientNet, Python* [🔗](#)

- Captured and curated a custom geo-localization dataset by photographing 15 campus regions across three distinct time periods, then trained models to predict GPS coordinates and camera angle from images.
- Evaluated ResNet, EfficientNet, and ConvNeXt architectures on multi-task regression and classification, applying robust preprocessing to exceed 90% accuracy across all classification tasks.

Distributed Network File System | *C, POSIX Threads, TCP/UDP Sockets, Distributed Systems* [🔗](#)

- Built a multithreaded distributed file system via TCP/IP (Naming and Storage Servers), utilizing POSIX threads and reader-writer locks to manage concurrent client access and enable reliable audio file streaming.
- Cut file-lookup latency via Trie-based routing and an LRU cache, and ensured high availability through 3x automatic replication with sub-second server failover.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL, R, LaTeX

ML & LLM Frameworks: PyTorch, Hugging Face Transformers, YOLOv8, FAISS, OpenAI API

Systems & Tools: Git, Linux, Docker, MongoDB

Web: React.js, Node.js, Express.js, FastAPI, Flask, Django, Tailwind CSS

Achievements

- Best Hardware Project in HackIIIT-2026.
- Top 10 Finalist, Lloyds IP & I Hackathon on Agentic AI (2025).
- Top 5 Finalist, Megathon 2025 – Chubb Problem Statement.